

Lab 14: Mobile_App_Investigations_Messaging

What You Need for this lab

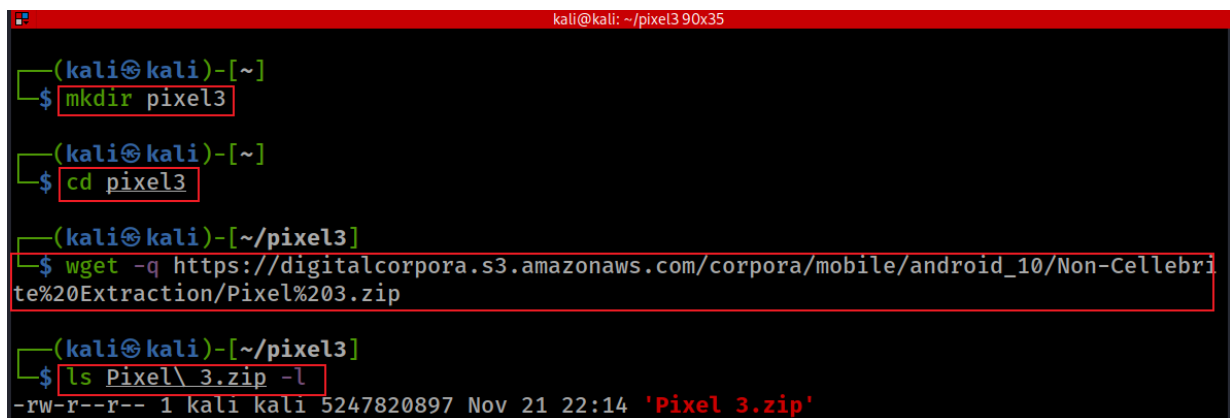
- Install Virtualbox : <https://www.virtualbox.org/wiki/Downloads>
- Install Kali 2021.4. : <https://old.kali.org/kali-images/kali-2021.4/>

Notes: Suggest You configure the disk size of Kali VM 80G because the size of each leakage cases image is 30G+

- Android Open-Source Project Apps :
 - https://digitalcorpora.s3.amazonaws.com/corpora/mobile/android_10/Non-Celebrite%20Extraction/Pixel%203.zip

Step 1

- Download Pixel 3 image



```
kali@kali: ~/pixel3 90x35
(kali㉿kali)-[~]
$ mkdir pixel3
(kali㉿kali)-[~]
$ cd pixel3
(kali㉿kali)-[~/pixel3]
$ wget -q https://digitalcorpora.s3.amazonaws.com/corpora/mobile/android_10/Non-Celebrite%20Extraction/Pixel%203.zip
(kali㉿kali)-[~/pixel3]
$ ls Pixel\ 3.zip -l
-rw-r--r-- 1 kali kali 5247820897 Nov 21 22:14 'Pixel 3.zip'
```

```
hainguyen@hainguyen: ~/pix × + v
Microsoft Windows [Version 10.0.26100.4652]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>ssh hainguyen@192.168.1.20
hainguyen@192.168.1.20's password:
Linux hainguyen 6.12.25-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.25-1kali1 (20
25-04-30) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Jul 13 11:30:04 2025 from 192.168.1.152
(hainguyen@hainguyen)-[~]
$ mkdir pixel3

(hainguyen@hainguyen)-[~]
$ cd pixel3

(hainguyen@hainguyen)-[~/pixel3]
$ wget -q https://digitalcorpora.s3.amazonaws.com/corpora/mobile/android_1
0/Non-Cellebrite%20Extraction/Pixel%203.zip

(hainguyen@hainguyen)-[~/pixel3]
$ ls Pixel\ 3.zip -l
-rw-rw-r-- 1 hainguyen hainguyen 5247820897 Nov 21 2020 'Pixel 3.zip'

(hainguyen@hainguyen)-[~/pixel3]
$ |
```

Verify hashes

hashdeep Pixel\ 3.zip

```
kali@kali: ~/pixel3 90x35
(kali@kali)-[~/pixel3]
$ hashdeep Pixel\ 3.zip
%%%% HASHDEEP-1.0
%% size,md5,sha256,filename
## Invoked from: /home/kali/pixel3
## $ hashdeep Pixel 3.zip
##
5247820897,9cc37ebb4e918ee5427de1fe1deecc,ca6918ef8b20486b6a5ded15609ac51318f377829480f9
3be3ba15364a8aa00a,/home/kali/pixel3/Pixel 3.zip
```

```
(hainguyen@hainguyen)-[~/pixel3]
$ hashdeep Pixel\ 3.zip
%%% HASHDEEP-1.0
%%% size,md5,sha256,filename
## Invoked from: /home/hainguyen/pixel3
## $ hashdeep Pixel 3.zip
##
5247820897,9cc37ebbbc4e918ee5427de1fe1deecc,ca6918ef8b20486b6a5ded15609ac513
18f377829480f93be3ba15364a8aa00a,/home/hainguyen/pixel3/Pixel 3.zip

(hainguyen@hainguyen)-[~/pixel3]
$ |
```

- unzip to get Pixel 3 image

```
kali@kali: ~/pixel3 90x35

(kali@kali)-[~/pixel3]
$ unzip -q Pixel\ 3.zip

(kali@kali)-[~/pixel3]
$ ls -l
total 5124836
drwxr-xr-x 11 kali kali      4096 Feb 15  2020 'Pixel 3'
-rw-r--r--  1 kali kali 5247820897 Nov 21 22:14 'Pixel 3.zip'
```

```
(hainguyen@hainguyen)-[~/pixel3]
$ unzip -q Pixel\ 3.zip

(hainguyen@hainguyen)-[~/pixel3]
$ ls -l
total 5124836
drwxrwxr-x 11 hainguyen hainguyen      4096 Feb 15  2020 'Pixel 3'
-rw-rw-r--  1 hainguyen hainguyen 5247820897 Nov 21  2020 'Pixel 3.zip'

(hainguyen@hainguyen)-[~/pixel3]
$ |
```

ls 'Pixel 3/data/data/'

```
(kali㉿kali)-[~/pixel3]
$ ls 'Pixel 3/data/data/'
com.android.chrome
com.android.hotwordenrollment.okgoogle
com.android.hotwordenrollment.xgoogle
com.android.keychain
com.android.mtp
com.android.nfc
com.android.providers.calendar
com.android.providers.contacts
com.android.providers.downloads
com.android.providers.media
com.android.providers.telephony
com.android.providers.userdictionary
com.android.service.ims.presence
com.android.traceur
com.android.vending
com.discord
com.enflick.android.TextNow
com.facebook.orca
com.google.android.apps.chromecast.app
com.google.android.apps.docs
com.google.android.apps.docs.editors.docs
com.google.android.apps.dreamliner
com.google.android.apps.enterprise.dmagent
com.google.android.apps.gcs
com.google.android.apps.helptrc
com.google.android.apps.magazines
com.google.android.apps.maps
com.google.android.apps.messaging
com.google.android.apps.nexuslauncher
com.google.android.apps.photos
com.google.android.apps.pixelmigrate
com.google.android.apps.safetyhub
com.google.android.apps.tachyon
com.google.android.apps.tips
com.google.android.apps.turbo
com.google.android.apps.tycho
com.google.android.apps.walletnfcrel
com.google.android.apps.wallpaper
com.google.android.apps.wellbeing
com.google.android.apps.work.oobconfig
com.google.android.apps.youtube.music
com.google.android.as
com.google.android.calendar
com.google.android.configupdater
com.google.android.contacts
com.google.android.deskclock
com.google.android.dialer
com.google.android.documentsui
com.google.android.ext.services
com.google.android.gm
com.google.android.gms
com.google.android.GoogleCamera
com.google.android.googlequicksearchbox
com.google.android.gsf
com.google.android.ims
com.google.android.inputmethod.latin
com.google.android.keep
com.google.android.markup
com.google.android.music
com.google.android.onetimeinitializer
com.google.android.partnersetup
com.google.android.pixel.setupwizard
com.google.android.projection.gearhead
com.google.android.settings.intelligence
com.google.android.setupwizard
com.google.android.syncadapters.contacts
com.google.android.tts
com.google.android.videos
com.google.android.webview
com.google.android.wfcactivation
com.google.android.youtube
com.google.ar.core
com.google.intelligence.sense
com.google.vr.apps.ornament
com.google.vr.vrcore
com.imgur.mobile
com.imo.android.imous
com.instagram.android
com.instagram.threadsapp
com.mewe
com.mywickr.wickr2
com.qti.ltebc
com.radicalapps.cyberdust
com.silentcircle.silentphone
com.skout.android
com.skype.raider
com.snapchat.android
com.spotify.music
com.thinkyeah.galleryvault
com.topjohnwu.magisk
com.twitter.android
com.venmo
com.viber.voip
com.whatsapp
com.wire
com.zhiliaoapp.musically
jp.naver.line.android
kik.android
org.telegram.messenger
org.thoughtcrime.securesms
org.torproject.torbrowser
```

```
hainguyen@hainguyen: ~/pix
(hainguyen㉿hainguyen)-[~/pixel3]
$ ls 'Pixel 3/data/data/'
com.android.chrome
com.android.hotwordenrollment.okgoogle
com.android.hotwordenrollment.xgoogle
com.android.keychain
com.android.mtp
com.android.nfc
com.android.providers.calendar
com.android.providers.contacts
com.android.providers.downloads
com.android.providers.media
com.android.providers.telephony
com.android.providers.userdictionary
com.android.service.ims.presence
com.android.traceur
com.android.vending
com.discord
com.enflick.android.TextNow
com.facebook.orca
com.google.android.apps.chromecast.app
com.google.android.apps.docs
com.google.android.apps.docs.editors.docs
com.google.android.apps.dreamliner
com.google.android.apps.enterprise.dmagent
com.google.android.apps.gcs
com.google.android.apps.helptrc
com.google.android.apps.magazines
com.google.android.apps.maps
com.google.android.apps.messaging
com.google.android.apps.nexuslauncher
com.google.android.apps.photos
com.google.android.apps.pixelmigrate
com.google.android.apps.safetyhub
com.google.android.apps.tachyon
com.google.android.apps.tips
com.google.android.apps.turbo
com.google.android.apps.tycho
com.google.android.apps.walletnfcrel
com.google.android.apps.wallpaper
com.google.android.apps.wellbeing
com.google.android.apps.work.oobconfig
com.google.android.apps.youtube.music
com.google.android.as
```

```
hainguyen@hainguyen: ~/pix
com.google.android.partnersetup
com.google.android.pixel.setupwizard
com.google.android.projection.gearhead
com.google.android.settings.intelligence
com.google.android.setupwizard
com.google.android.syncadapters.contacts
com.google.android.tts
com.google.android.videos
com.google.android.webview
com.google.android.wfcactivation
com.google.android.youtube
com.google.ar.core
com.google.intelligence.sense
com.google.vr.apps.ornament
com.google.vr.vrcore
com.igur.mobile
com.imo.android.imoos
com.instagram.android
com.instagram.threadsapp
com.mewe
com.nywicker.wicker2
com.qti.ltebc
com.radicalapps.cyberdust
com.silentcircle.silentphone
com.skout.android
com.skype.raider
com.snapchat.android
com.spotify.music
com.thinkyeah.galleryvault
com.topjohnwu.magisk
com.twitter.android
com.venmo
com.viber.voip
com.whatsapp
com.wire
com.zhiliaoapp.musically
jp.naver.line.android
kik.android
org.telegram.messenger
org.thoughtcrime.securesms
org.torproject.torbrowser

(hainguyen@hainguyen)-[~/pixel3]
$
```

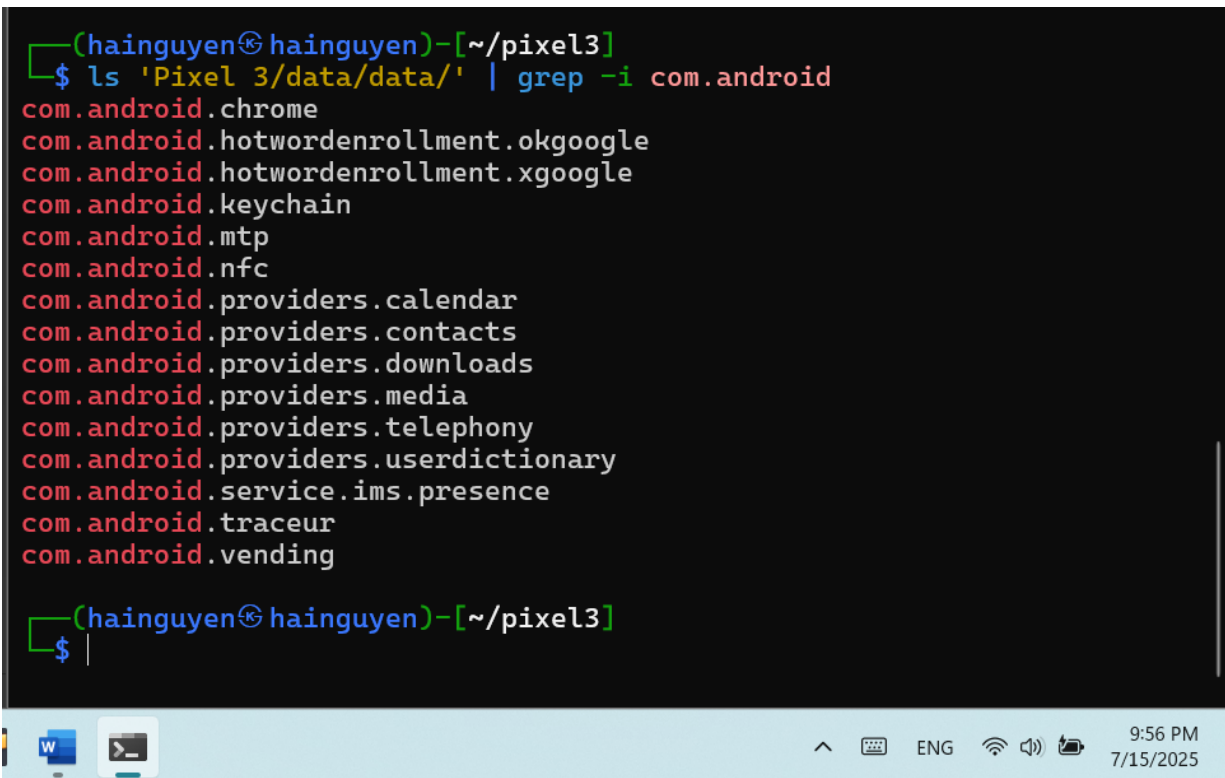
Step 2. Investigating Messaging Services

- Show all AOSP Apps/Packages: *com.android.**

```
(kali@kali)-[~/pixel3]
$ ls 'Pixel 3/data/data/' | grep -i com.android
com.android.chrome
com.android.hotwordenrollment.okgoogle
com.android.hotwordenrollment.xgoogle
com.android.keychain
com.android.mtp
com.android.nfc
com.android.providers.calendar
com.android.providers.contacts
com.android.providers.downloads
com.android.providers.media
com.android.providers.telephony
com.android.providers.userdictionary
com.android.service.ims.presence
com.android.traceur
com.android.vending
```

```
(hainguyen@hainguyen)-[~/pixel3]
$ ls 'Pixel 3/data/data/' | grep -i com.android
com.android.chrome
com.android.hotwordenrollment.okgoogle
com.android.hotwordenrollment.xgoogle
com.android.keychain
com.android.mtp
com.android.nfc
com.android.providers.calendar
com.android.providers.contacts
com.android.providers.downloads
com.android.providers.media
com.android.providers.telephony
com.android.providers.userdictionary
com.android.service.ims.presence
com.android.traceur
com.android.vending

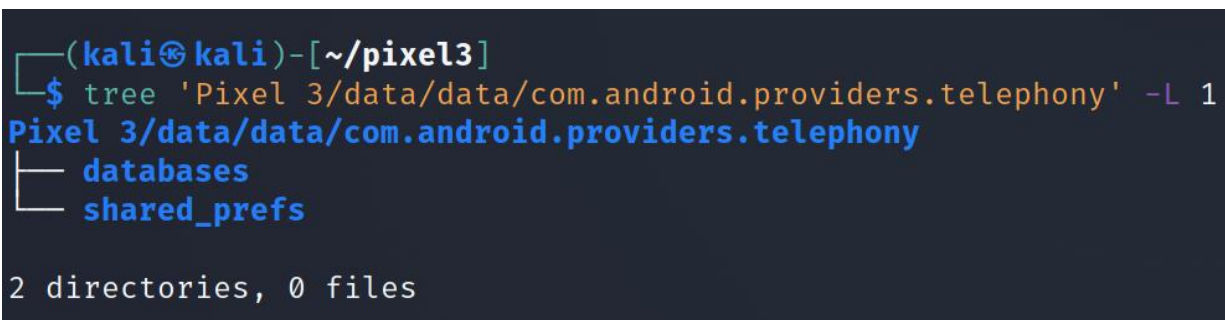
(hainguyen@hainguyen)-[~/pixel3]
$ |
```



- Find the location of Java package

```
(kali@kali)-[~/pixel3]
$ tree 'Pixel 3/data/data/com.android.providers.telephony' -L 1
Pixel 3/data/data/com.android.providers.telephony
├── databases
└── shared_prefs

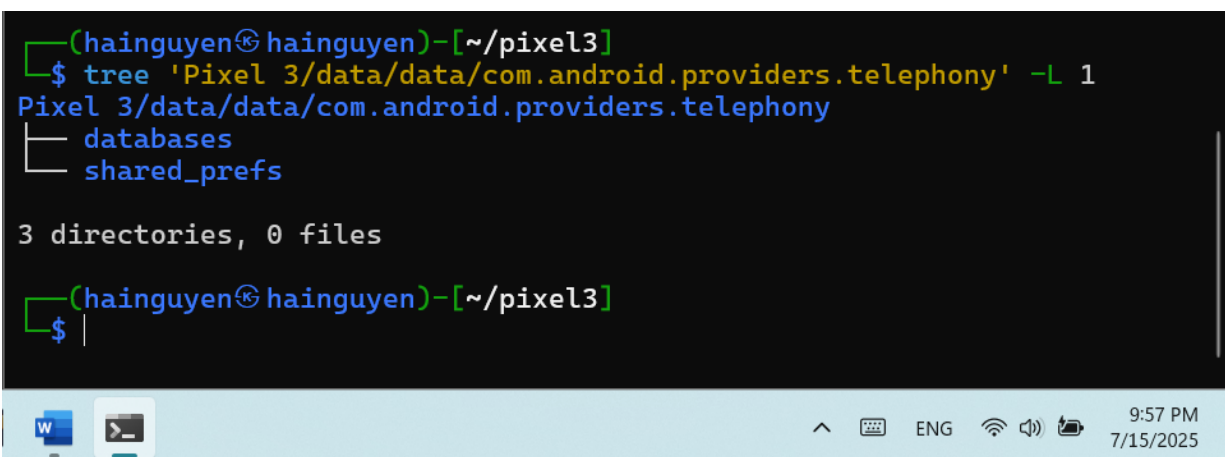
2 directories, 0 files
```



```
(hainguyen@hainguyen)-[~/pixel3]
$ tree 'Pixel 3/data/data/com.android.providers.telephony' -L 1
Pixel 3/data/data/com.android.providers.telephony
├── databases
└── shared_prefs

3 directories, 0 files

(hainguyen@hainguyen)-[~/pixel3]
$ |
```



- Find the database location

```
(kali@kali)-[~/pixel3]
$ tree 'Pixel 3/data/data/com.android.providers.telephony/databases'
Pixel 3/data/data/com.android.providers.telephony/databases
├── mmssms.db

0 directories, 1 file
```

```
(hainguyen@hainguyen)-[~/pixel3]
$ tree 'Pixel 3/data/data/com.android.providers.telephony/databases'

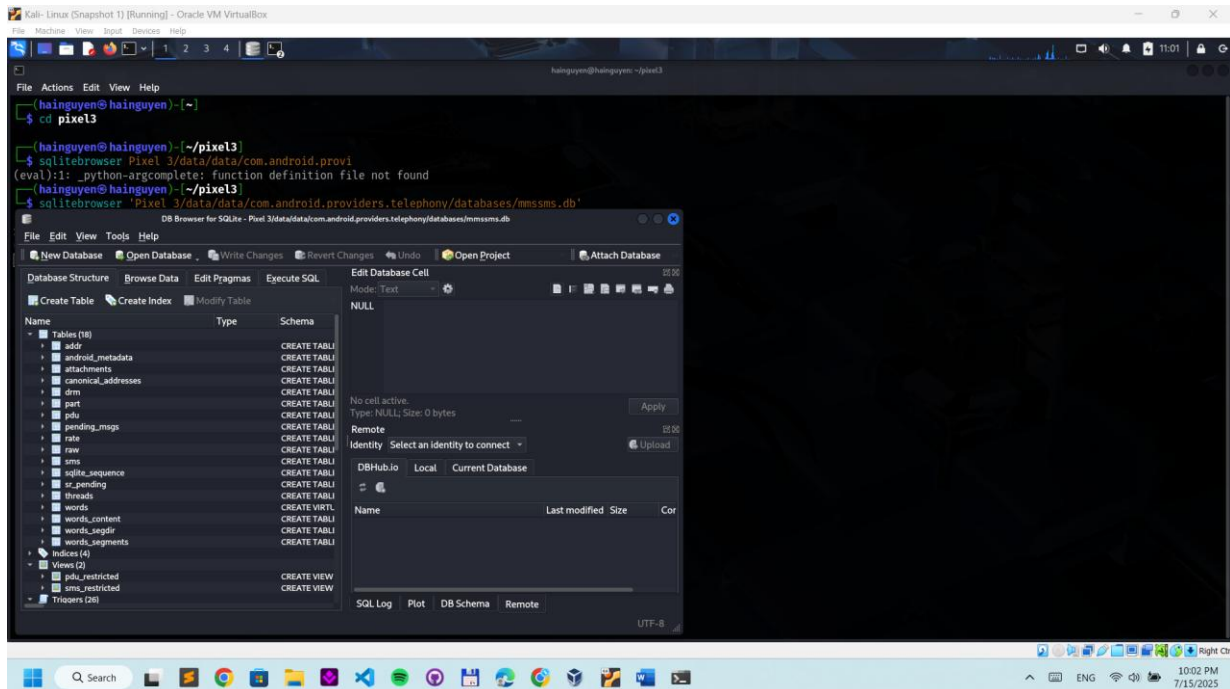
Pixel 3/data/data/com.android.providers.telephony/databases
├── mmssms.db

1 directory, 1 file

(hainguyen@hainguyen)-[~/pixel3]
$ |
```

- Show all tables in the database

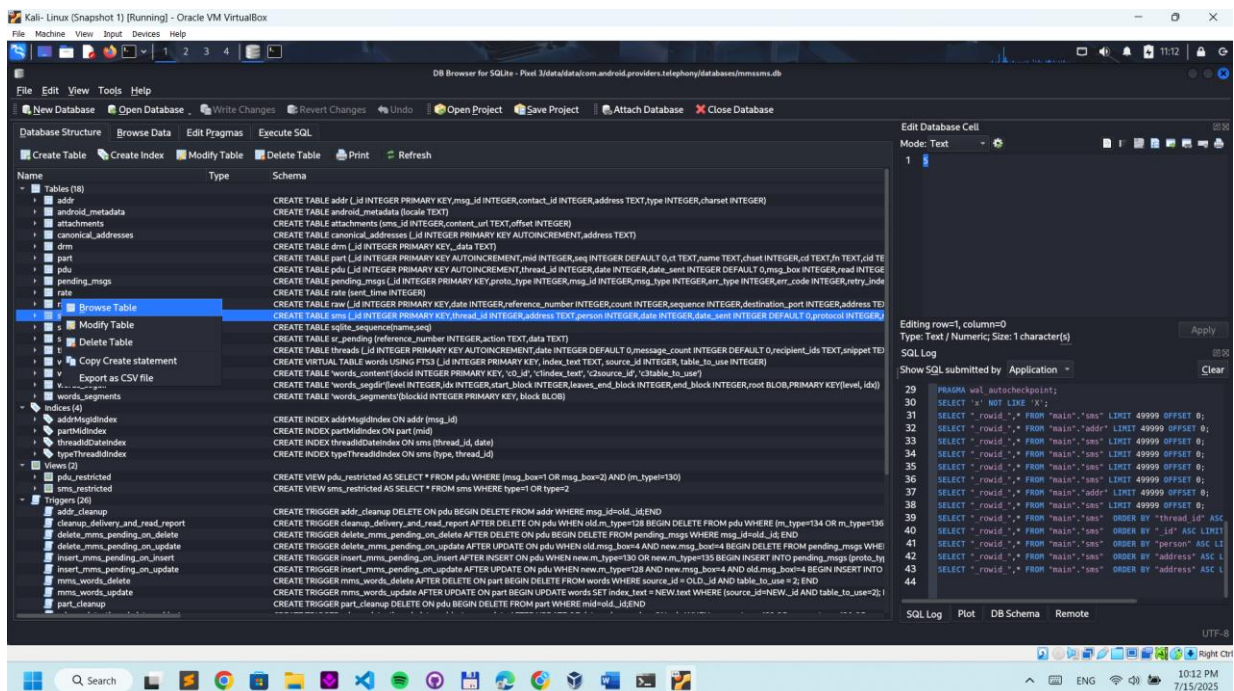
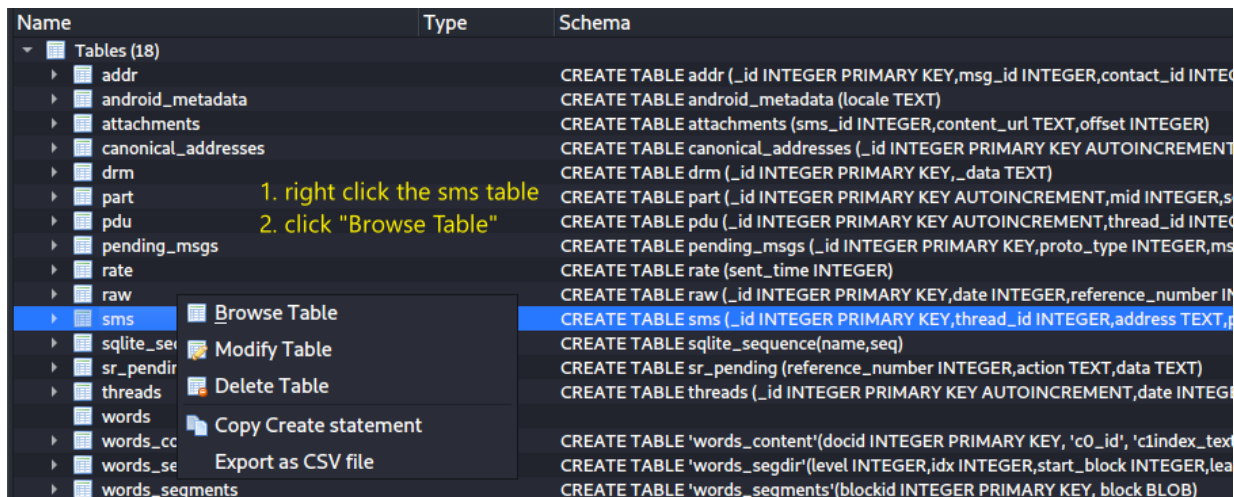
```
(hainguyen@hainguyen)-[~/pixel3]
$ sqlitebrowser 'Pixel 3/data/data/com.android.providers.telephony/databases/mmssms.db'
```



The screenshot shows the SQLite Browser application interface. The left pane displays the database structure with a tree view showing tables and views. The right pane shows the 'Edit Database Cell' dialog with a NULL value. The bottom status bar indicates the system time as 10:02 PM on 7/15/2025.

- *sms*: message log
 - Only system, phone or the default message app can have full access of sms data
- *sms_restricted*:
 - Only contains sent or received messages.

- *threads*
 - Group all messages based on incoming phone numbers and time period?
- Exam *sms* table structure



- Find clues to answer the question

Kali - Linux (Snapshot 1) [Running] - Oracle VM VirtualBox

DB Browser for SQLite - Print 31data\data\com.android.providers.telephony\databases\mmssms.db

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database

Database Structure Browse Data Edit Pragma Execute SQL

addr sms

Table: sms

id	thread_id	address	person	date	date_sent	protocol	read	status	type	reply_path_present	subject	body	service_center	locked	sub_id	error_code
1	2	4 +1323205525	NULL	1580322045251	1580322043000	0	0	-1	1	0	NULL	<- Please enter 304868 into LINE ...	6502531234	0	1	0 com.goog
2	3	5 732873	NULL	1580322046631	1580322045000	0	0	-1	1	0	NULL	Telegram code: 59389...	6502531234	0	1	0 com.goog
3	4	6 +18437514985	NULL	1580322069689	1580322069000	0	0	-1	1	0	NULL	<-> Your Signal verification code: ...	6502531234	0	1	0 com.goog
4	5	7 +12253690855	NULL	1580322728551	1580322727000	0	0	-1	1	0	NULL	Your imo code is 685614...	6502531234	0	1	0 com.goog
5	6	8 +12347570651	NULL	1580322820350	1580322819000	0	0	-1	1	0	NULL	Your Skout verification code is 2661...	6502531234	0	1	0 com.goog
6	7	9 +12349000124	NULL	1580322903320	1580322902000	0	0	-1	1	0	NULL	Your Skout verification code is 9073...	6502531234	0	1	0 com.goog
7	8	10 47543	NULL	1580323020883	1580323027000	0	0	-1	1	0	NULL	<-> Your WhatsApp code: 362-593...	6502531234	0	1	0 com.goog
8	9	5 732873	NULL	1580323209115	1580323207000	0	0	-1	1	0	NULL	Use 3649 as Microsoft account ...	6502531234	0	1	0 com.goog
9	10	11 59109	NULL	1580323308596	1580323307000	0	0	-1	1	0	NULL	[e][TikTok] 6327 is your verificatio...	6502531234	0	1	0 com.goog
10	11	12 837482	NULL	1580323813826	1580323812000	0	0	-1	1	0	NULL	FB-55286 is ThisIs's Facebook ...	6502531234	0	1	0 com.goog
11	12	13 32665	NULL	1580323824938	1580323824000	0	0	-1	1	0	NULL	Confirmed! To edit SMS preferences g...	6502531234	0	1	0 com.goog
12	13	5 732873	NULL	1580323897037	1580323896000	0	0	-1	1	0	NULL	<-> Your Inqur verification code is ...	6502531234	0	1	0 com.goog
13	14	14 44398	NULL	1580323903357	1580323902000	0	0	-1	1	0	NULL	369967 is your verification code for...	6502531234	0	1	0 com.goog
14	15	15 +17162394257	NULL	1580518003531	1580518001000	0	1	-1	1	0	NULL	Your Wickr Me confirmation code is ...	6502531234	0	1	0 com.goog
15	16	16 +13232080934	NULL	1580519052798	1580519051000	0	1	-1	1	0	NULL	Your Wire code is 172488...	6502531234	0	1	0 com.goog
16	17	13 32665	NULL	158057526393	158057524000	0	0	-1	1	0	NULL	Xavier Mccollum is a new Facebook...	6502531234	0	1	0 com.goog
17	18	13 32665	NULL	1580919741561	1580919738000	0	0	-1	1	0	NULL	▲ ThisIs, you have 1 new ...	6502531234	0	1	0 com.goog
18	19	13 32665	NULL	1581063359840	1581063357000	0	0	-1	1	0	NULL	▲ ThisIs, you have 1 new ...	6502531234	0	1	0 com.goog
19	20	13 32665	NULL	1581185821854	1581185818000	0	0	-1	1	0	NULL	▲ ThisIs, you have 1 new ...	6502531234	0	1	0 com.goog
20	21	17 9195790479	NULL	1581273702392	0	NULL	1	-1	2	0	NULL	I am on my way to the coffee shop. D...	NULL	0	2	0 com.goog
21	22	13 32665	NULL	1581444160908	1581444159000	0	0	-1	1	0	NULL	▲ ThisIs, you have 6 new ...	6502531234	0	2	0 com.goog

Go to: 1

SQL Log Plot DB Schema Remote UTF-8

- Which message was received on Jan 29, 2020 around 6:20 pm?

<https://www.epochconverter.com/>

Time zone converter

Timestamp list

LDAP converter

WebKRI/Chrome timestamp

Unix hex timestamp

Cocoa Core Data timestamp

Mac HFS+ timestamp

SAS timestamp

Seconds/days since year 0

Bin/Oct/Hex converter

Countdown in seconds

Epoch clock

Date and Time

Week numbers

Weeks by year

Day numbers

Days by year

Years/leap years

Calculate the difference between two dates

Programming

More

Comments & questions

Yr Mon Day Hr Min Sec

2020 1 29 06 20 00 PM GMT Human date to Timestamp

Epoch timestamp: 1580322000

Timestamp in milliseconds: 1580322000000

Date and time (GMT): Wednesday, January 29, 2020 6:20:00 PM

Date and time (your time zone): Wednesday, January 29, 2020 1:20:00 PM GMT-05:00

- Who sent the message?

```
(hainguyen@hainguyen)-[~/pixel3]
$ date -d '@1580322045' -u
Wed Jan 29 06:20:45 PM UTC 2020
(hainguyen@hainguyen)-[~/pixel3]
$
```


Where to find sender's phone number?

Exam *canonical_addresses* table

Table: canonical_addresses

	_id	address
1	4	+13233205525
2	5	732873
3	6	+18437514985
4	7	+12253690855
5	8	+12347570651
6	9	+12349008124
7	10	47543
8	11	59109
9	12	837402

thread/phone ID phone number

DB Browser for SQLite - Print 3\data\data\com.android.providers.telephony\databases\mmssms.db

Table: canonical_addresses

	_id	address
1	4	+13233205525
2	5	732873
3	6	+18437514985
4	7	+12253690855
5	8	+12347570651
6	9	+12349008124
7	10	47543
8	11	59109
9	12	837402
10	13	32665
11	14	44398
12	15	+17162394257
13	16	+13233208934
14	17	+19195790479
15	18	'WAP_PUSH_SII'
16	19	Service messages
17	20	22800

Which table contains MMS information ?

Table: part

	_id	mid	seq	ct	name	chset	cd	fn	cid	cl	ctt_s	ctt_t	_data	text
1	1	1	-1	application/smil	NULL	NULL	NULL	NULL	<smil>	smil.xml	NULL	NULL	NULL	<smil> <head> <layout> <root-layout/...
2	2	1	0	text/plain	NULL	106	NULL	NULL	<text0...	text00...	NULL	NULL	NULL	Here's Signal.
3	3	2	0	text/plain	body	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	It looks like the messaging is run through the...
4	4	3	-1	application/smil	NULL	NULL	NULL	NULL	<smil>	smil.xml	NULL	NULL	NULL	<smil> <head> <layout> <root-layout/...
5	5	3	0	text/plain	NULL	106	NULL	NULL	<text0...	text00...	NULL	NULL	NULL	It might be an attempt at unified messaging...?

_id	mid	seq	ct	name	chset	cd	fn	cid	ct_s	ct_t	ct_data	text
1	1	-1	application/smil	NUL	NUL	NUL	<smil>	smil.xml	NUL	NUL	NUL	<smil>=<layout>=<root>=<layout>=
2	1	0	text/plain	NUL	106	NUL	<text0000001>	text0000001.txt	NUL	NUL	NUL	Here's Signal.
3	2	0	text/plain	body	NUL	NUL	NUL	NUL	NUL	NUL	NUL	It looks like the messaging is run ..
4	3	-1	application/smil	NUL	NUL	NUL	<smil>	smil.xml	NUL	NUL	NUL	<smil>=<head>=<layout>=<root>=<layout>=
5	3	0	text/plain	NUL	106	NUL	<text0000001>	text0000001.txt	NUL	NUL	NUL	It might be an attempt at unified ..
6	4	0	text/plain	body	NUL	NUL	NUL	NUL	NUL	NUL	NUL	Maybe?
7	5	-1	application/smil	NUL	NUL	NUL	<smil>	smil.xml	NUL	NUL	NUL	<smil>=<head>=<layout>=<root>=<layout>=
8	5	0	text/plain	NUL	106	NUL	<text0000001>	text0000001.txt	NUL	NUL	NUL	It will be interesting to see where ..
9	6	0	text/plain	body	NUL	NUL	NUL	NUL	NUL	NUL	NUL	No, thank you
10	7	0	text/plain	body	NUL	NUL	NUL	NUL	NUL	NUL	NUL	That is interesting.

Part table stores below information:

- **_id**: Distinguish between different attachments of a multimedia message (including text, picture, audio, video and other formats)
- **ct**: indicates what type of attachment
- **data**: indicate where the attachment is stored on the phone
- **text**: Represents the body of the MMS.

YOU MUST SUBMIT A FULL-SCREEN IMAGE FOR FULL CREDIT!

Save the document with the filename "**YOUR NAME Lab 14.pdf**", replacing "YOUR NAME" with your real name.

Email the image to the instructor as an attachment to an e-mail message. Send it to: **xxx@fe.edu.vn** with a subject line of "**Lab 14 From YOUR NAME**", replacing "YOUR NAME" with your real name.